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In these resonance structures, the (+) charge residues is present on the ring structure. It is quite evident that resonance delocalizes the positive charge on the atoms of the ring, making the ring less electron rich than benzene. Here overall density of benzene ring decreases due to –I effect of –CHO group there by deactivating the benzene for electrophilic attack. However resonating structure shows that electron density is more in meta position. Compared to o & p-position. Hence –CHO group is a meta-director and deactivator.

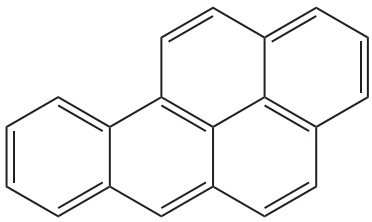
Evaluate Yourself

25) Toluene undergoes nitration easily than benzene.Why?

13.5.2. Carcinogenity and toxicity

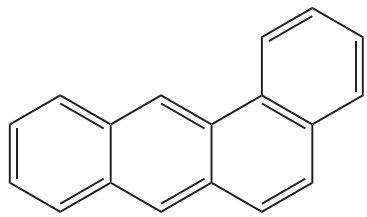
Benzene and polycyclic aromatic hydrocarbons (PAH) are ubiquitous environmental pollutants generated during incomplete combustion of coal oil, petrol and wood. Some PAH originate from open burning, natural seepage of petroleum and coal deposits and volcanic activities. They are toxic, mutagenic and carcinogenic. It has hematological. immunological and neurological effect on humans. They are radiomimetic and prolonged exposure leads to genetic damage.Some of the examples of PAH are.

"L" shaped polynuclear hydrocarbons are much more toxic & carcinogenic



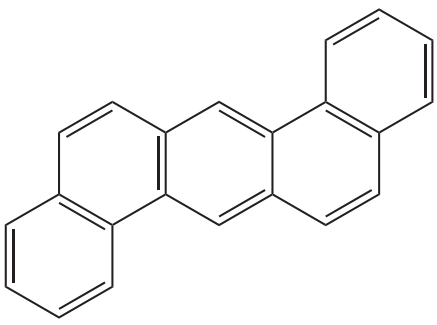
1,3- Benzpyrene

Found in cigarette smoke



1,3- Benzanthracine

Found in tobacco and cigarette and charcoal boiled food



1,2,5,6 dibenzanthracene

Found in gasoline exhaust and barbaccued food